

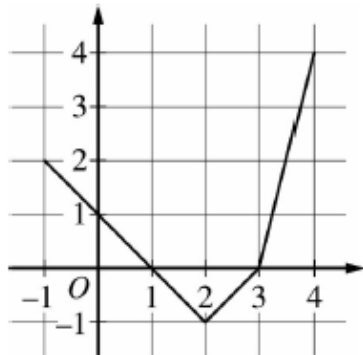
“The problem sets you begin getting a few weeks in are very beneficial. By the time you reach the AP test, you will feel very familiar and comfortable with the format, as long as you did them each week.” A 2019 Nerd

FR1. 1999 – BC 6 (Calculator)

Let $y = f(x)$ be the function whose graph goes through the point $(3, 6)$ and whose derivative is given by $\frac{dy}{dx} = \frac{1+e^x}{x^2}$.

- Write an equation of the line tangent to the graph of f at $x = 3$ and use it to approximate $f(3.2)$.
- Use Euler's method, starting at $x = 3$, with a step size of 0.1, to approximate $f(3.2)$. Use f'' to explain why this approximation is less than $f(3.2)$.
- Use $\int_3^{3.2} f'(x) dx$ to evaluate $f(3.2)$.

FR2. Practice 2017 Test (No Calculator)

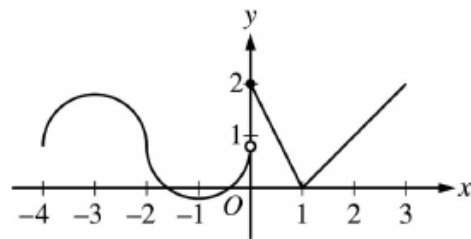
Graph of f

Let f be a continuous function defined on the closed interval $-1 \leq x \leq 4$. The graph of f , consisting of three line segments, is shown above. Let g be the function defined by

$$g(x) = 5 + \int_2^x f(t) dt \text{ for } -1 \leq x \leq 4.$$

- Find $g(4)$.
- On what intervals is g increasing? Justify your answer.
- On the closed interval $-1 \leq x \leq 4$, find the absolute minimum value of g and find the absolute maximum value of g . Justify your answers.
- Let $h(x) = x \cdot g(x)$. Find $h'(2)$.

MC1.



Graph of f

The graph of the piecewise-defined function f is shown in the figure above. The graph has a vertical tangent line at $x = -2$ and horizontal tangent lines at $x = -3$ and $x = -1$. What are all values of x , $-4 < x < 3$, at which f is continuous but not differentiable?

- (A) $x = 1$
- (B) $x = -2$ and $x = 0$
- (C) $x = -2$ and $x = 1$
- (D) $x = 0$ and $x = 1$

MC2. $\int_1^e \left(\frac{x^2 - 1}{x} \right) dx =$

- (A) $e - \frac{1}{e}$
- (B) $e^2 - e$
- (C) $\frac{e^2}{2} - e + \frac{1}{2}$
- (D) $\frac{e^2}{2} - \frac{3}{2}$

MC3. $\frac{d}{dt}(\arcsin 2x) =$

- (A) $\frac{1}{\sqrt{1 - 4x^2}}$
- (B) $\frac{2}{\sqrt{1 - 4x^2}}$
- (C) $\frac{2}{\sqrt{1 - 4x^2}} \frac{dx}{dt}$
- (D) $\frac{2}{1 + 4x^2} \frac{dx}{dt}$

MC4. CA A rain barrel collects water off the roof of a house during three hours of heavy rainfall. The height of the water in the barrel increases at the rate of $r(t) = 4t^3 e^{-1.5t}$ feet per hour, where t is the time in hours since the rain began. At time $t = 1$ hour, the height of the water is 0.75 foot. What is the height of the water in the barrel at time $t = 2$ hours?

- (A) 1.361 ft
- (B) 1.500 ft
- (C) 1.672 ft
- (D) 2.111 ft